

CS 271 Project Report

The Evolution of Image Generation: From Autoencoders to Diffusion

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Abstract

This project investigates three popular generative modeling approaches—Variational Autoencoders (VAE), Generative Adversarial Networks (GAN), and Denoising Diffusion Probabilistic Models (DDPM), for the task of unconditional image generation on the CIFAR-10 dataset. Each model was implemented and trained on the same dataset and preprocessing pipeline for a fair comparison. We evaluate their outputs using both qualitative visual inspection and quantitative metrics, including Inception Score (IS), Fréchet Inception Distance (FID), and Kernel Inception Distance (KID).

Our results show that the VAE produces recognizable but blurry outputs ($\text{FID} = 241.33$, $\text{IS} = 1.99 \pm 0.03$) due to its Gaussian decoder assumptions and reconstruction-based objective. GANs produce sharper images ($\text{FID} = 29.29$, $\text{IS} = 8.58 \pm 0.20$) but suffer from training instability and occasional mode collapse. Diffusion models generate the highest-quality images with strong detail and diversity ($\text{FID} = 11.03$, $\text{IS} = 8.58 \pm 0.20$), although at a significantly higher sampling cost. Overall, diffusion models achieve the best quantitative scores, followed by GAN, then VAE. This study highlights the trade-offs between sample fidelity, training stability, and computational cost in modern generative modeling.

1 Introduction

Generative modeling aims to learn the underlying distribution of data so that new, realistic samples can be synthesized. This capability is crucial in applications such as data augmentation, image synthesis, restoration, and creative AI. The ability to generate high-quality, diverse images has profound implications across computer vision, graphics, and artificial intelligence research.

In recent years, three major architectures have dominated the field of generative modeling:

- **Variational Autoencoders (VAEs)** - probabilistic, reconstruction-based models that learn a continuous latent representation [1].
- **Generative Adversarial Networks (GANs)** - adversarial, game-theoretic models that pit a generator against a discriminator [2].
- **Diffusion Models** - iterative noise-removal models that currently power state-of-the-art systems such as Stable Diffusion [3].

This project implements and compares these three fundamental approaches on the CIFAR-10 dataset, consisting of 32×32 natural images across 10 object categories. By training all models on the same dataset with consistent preprocessing, we provide a fair empirical comparison of their strengths and weaknesses. Our goal is to understand the trade-offs between image quality, training stability, computational requirements, and generation speed across these three paradigms.

2 Background and Related Work

2.1 Variational Autoencoders

Variational Autoencoders (VAEs), introduced by Kingma and Welling [1], model data via latent variables using variational inference. The key components are:

- **Encoder** — outputs parameters μ and $\log \sigma^2$ of an approximate posterior distribution $q_\phi(z|x)$
- **Reparameterization trick** — enables backpropagation through stochastic sampling: $z = \mu + \sigma \cdot \epsilon$, where $\epsilon \sim \mathcal{N}(0, 1)$
- **Decoder** — reconstructs the input image from the latent code: $p_\theta(x|z)$

- **Loss function** — combines reconstruction loss and KL divergence:

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - D_{KL}(q_\phi(z|x)||p(z)) \quad (1)$$

VAEs are stable to train and provide a meaningful latent space for interpolation and manipulation. However, they tend to produce blurry images due to the Gaussian assumptions in the decoder and the reconstruction-based objective that encourages averaging.

2.2 Generative Adversarial Networks

Generative Adversarial Networks (GANs), proposed by Goodfellow et al. [2], consist of two neural networks trained in opposition:

- **Generator (G)** — maps random noise vectors $z \sim p(z)$ to synthetic images $G(z)$
- **Discriminator (D)** — distinguishes between real images from the dataset and fake images from the generator

Training a GAN is formulated as a min-max game:

$$\min_G \max_D \mathbb{E}_{x \sim p_{data}}[\log D(x)] + \mathbb{E}_{z \sim p(z)}[\log(1 - D(G(z)))] \quad (2)$$

The DCGAN architecture [5] introduced convolutional layers, batch normalization, and specific activation functions that stabilize GAN training. GANs can produce sharp, photo-realistic images but suffer from training instability, mode collapse (where the generator produces limited variety), and difficulty in convergence [11].

2.3 Diffusion Models

Denoising Diffusion Probabilistic Models (DDPMs), developed by Sohl-Dickstein et al. [6] and significantly improved by Ho et al. [3], gradually corrupt images with Gaussian noise and learn to reverse this corruption process.

The **forward diffusion process** adds noise over T timesteps:

$$q(x_t|x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I) \quad (3)$$

where $\bar{\alpha}_t = \prod_{i=1}^t (1 - \beta_i)$ and β_t follows a noise schedule.

The **reverse process** learns to denoise:

- A U-Net [8] with time embeddings predicts the noise ϵ added at timestep t

- The training objective is:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t,x_0,\epsilon} [\|\epsilon - \epsilon_\theta(x_t, t)\|^2] \quad (4)$$

Sampling involves iteratively denoising from pure Gaussian noise over T steps. Diffusion models currently achieve state-of-the-art image generation quality [9, 10] and power systems like Stable Diffusion [7].

3 Dataset and Preprocessing

3.1 CIFAR-10 Dataset

The CIFAR-10 dataset [4] is a widely-used benchmark for image generation and classification tasks. It consists of:

- 60,000 color images ($32 \times 32 \times 3$ RGB)
- 10 object classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck
- 50,000 training images and 10,000 test images
- Balanced distribution with 6,000 images per class

3.2 Preprocessing Pipeline

All three models use identical preprocessing to ensure fair comparison:

- **Tensor conversion** — Images converted from PIL format to PyTorch tensors
- **Normalization** — Pixel values scaled from $[0, 255]$ to $[-1, 1]$ using:

$$x_{\text{normalized}} = \frac{x}{127.5} - 1 \quad (5)$$

This range is optimal for **tanh** activation functions used in all three generators/decoders.

3.3 DataLoader Configuration

- **VAE**: Batch size = 256, Workers = 2, Shuffle = True
- **GAN**: Batch size = 128, Workers = 2, Shuffle = True

- **Diffusion:** Batch size = 64, Workers = 2, Shuffle = True

The diffusion model uses a smaller batch size due to its larger U-Net architecture and higher memory requirements.

4 Methodology

4.1 VAE Architecture

Our VAE implementation follows the convolutional architecture design:

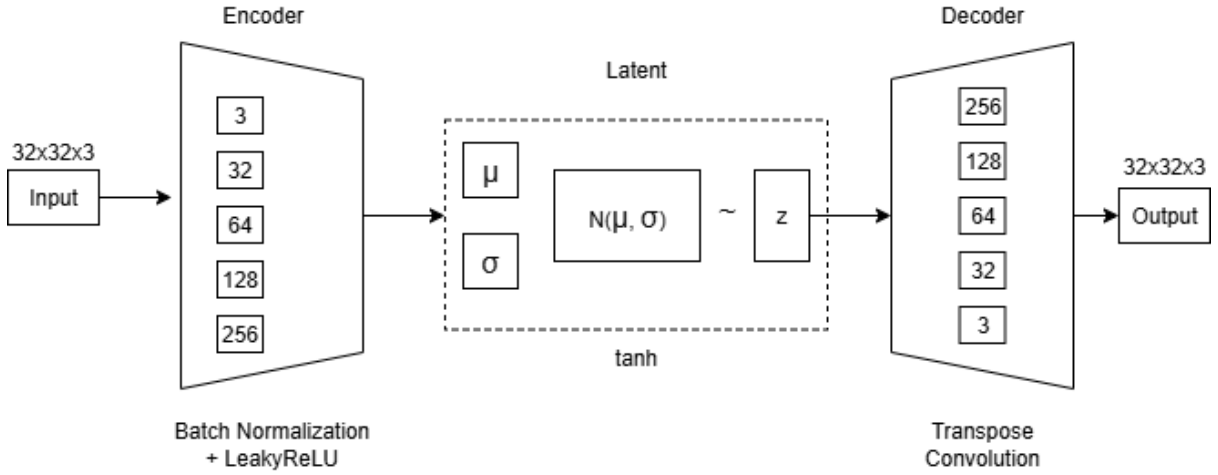


Figure 1: VAE Architecture Diagram

Encoder:

- 4 convolutional layers with stride 2: (32, 16, 8, 2) spatial dimensions
- Channel progression: $3 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256$
- Batch normalization and LeakyReLU(0.2) activation after each layer
- Flattened feature map: $256 \times 2 \times 2 = 1024$ features
- Two linear heads: $\mu, \log \sigma^2 \in \mathbb{R}^{256}$ (latent dimension = 256)

Decoder:

- Linear layer: $256 \rightarrow 1024$, reshaped to (256, 2, 2)
- 4 transposed convolutional layers with stride 2
- Channel progression: $256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 3$

- Batch normalization and ReLU activation (except final layer)
- Final activation: `tanh` for output in $[-1, 1]$

Loss Function:

$$\mathcal{L} = \text{MSE}(x, \hat{x}) + \beta \cdot D_{KL} \quad (6)$$

where MSE is the reconstruction loss and β is annealed from 0 to 1 over 10 warmup epochs to prevent posterior collapse [14].

4.2 GAN Architecture

Our GAN follows the DCGAN architecture [5]:

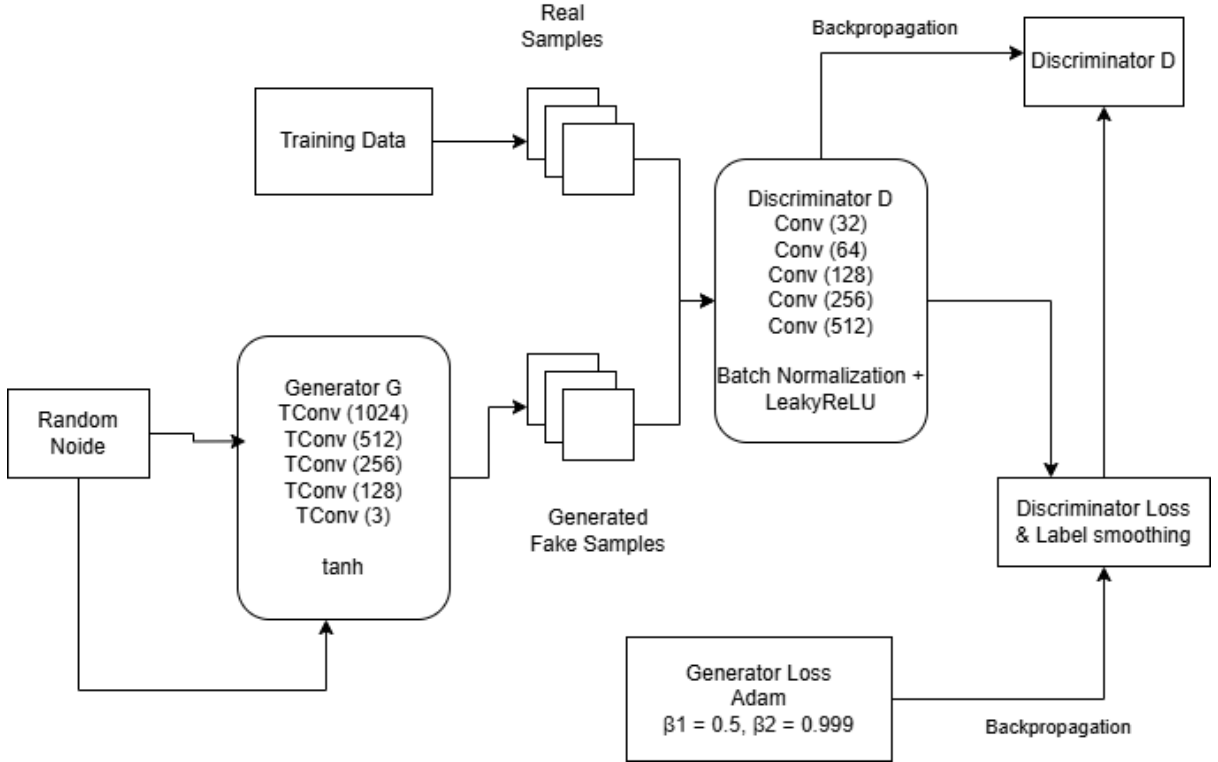


Figure 2: GAN Architecture Diagram

Generator:

- Input: Noise vector $z \in \mathbb{R}^{128}$ sampled from $\mathcal{N}(0, 1)$
- Project to $(128 \times 8, 4, 4)$ using transposed convolution
- 4 transposed convolutional blocks with stride 2
- Channel progression: $1024 \rightarrow 512 \rightarrow 256 \rightarrow 128 \rightarrow 3$

- Feature map sizes: 128 (base channels/feature maps)
- Batch normalization and ReLU activation (except final layer)
- Final activation: `tanh`

Discriminator:

- 4 convolutional blocks with stride 2
- Channel progression: $3 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$
- Batch normalization (not on first layer) and LeakyReLU(0.2)
- Final convolution: $(512, 4, 4) \rightarrow (1, 1, 1)$
- No sigmoid (using BCEWithLogitsLoss for stability)

Training Strategy:

- Binary cross-entropy loss with label smoothing
- Alternating gradient updates: 1 discriminator step per generator step
- Optimizer: Adam with $\beta_1 = 0.5$, $\beta_2 = 0.999$
- DCGAN weight initialization [5]

4.3 Diffusion Model (DDPM)

Our diffusion model implements the DDPM framework [3]:

U-Net Architecture:

- Base channels: 128 with channel multipliers $[1, 2, 4]$
- Three downsampling levels: $32 \rightarrow 16 \rightarrow 8$
- ResNet blocks: 2 per resolution level
- Self-attention layers at 16×16 and 8×8 resolutions (4 attention heads)
- Sinusoidal time embedding: 256 dimensions
- Skip connections between encoder and decoder
- GroupNorm normalization with adaptive group sizing
- SiLU (Swish) activations

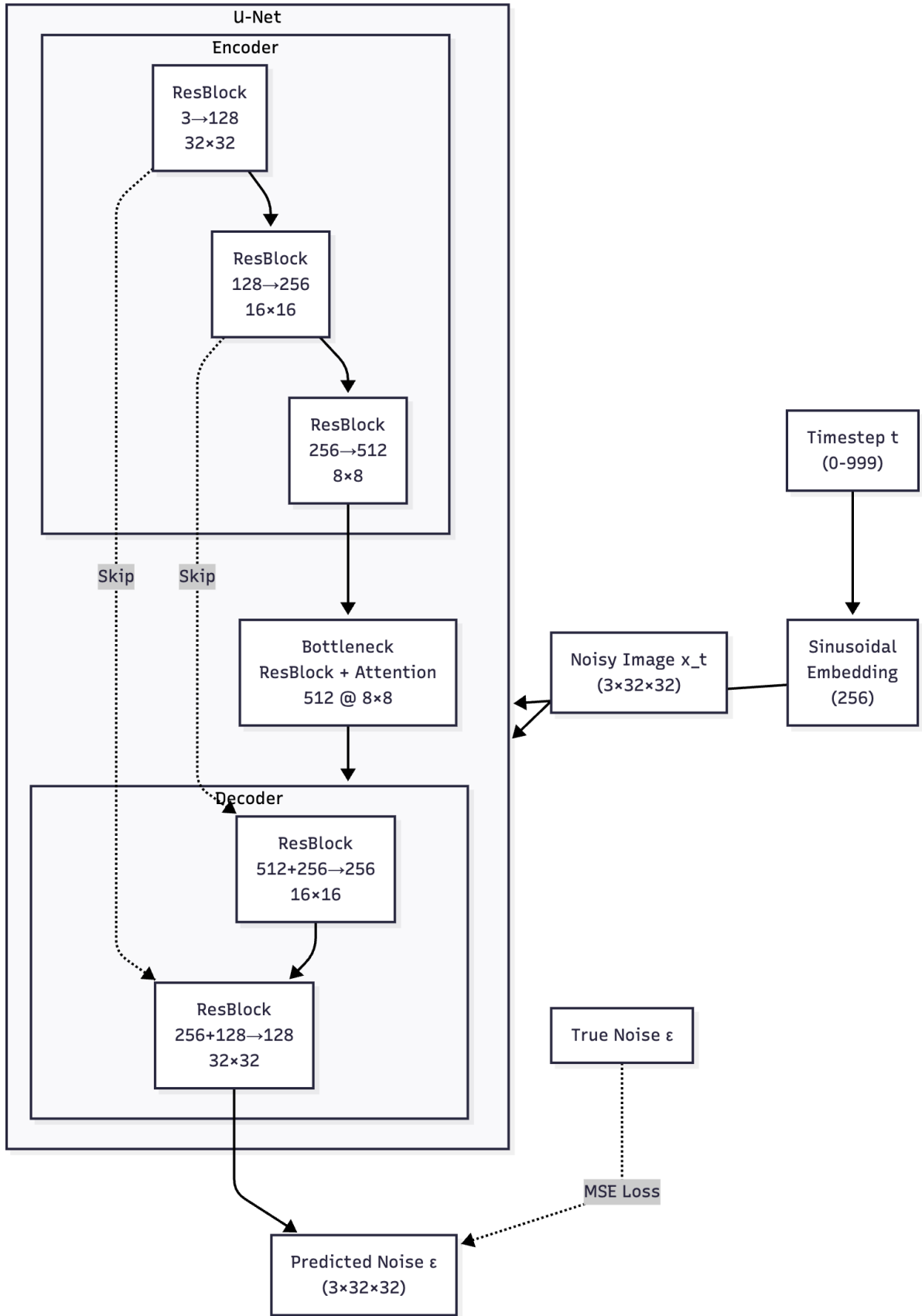


Figure 3: Diffusion Model Architecture Diagram

Noise Schedule:

- Cosine schedule [9] over $T = 1000$ timesteps
- Smoother noise addition compared to linear schedule
- Parameters: $s = 0.008$, $\beta \in [0.0001, 0.9999]$

Training:

- Loss: $\mathcal{L} = \|\epsilon - \epsilon_\theta(x_t, t)\|^2$ (simple MSE objective)
- Optimizer: AdamW with learning rate 2e-4
- Cosine learning rate schedule with 10-epoch warmup
- Gradient clipping: max norm = 1.0
- Exponential Moving Average (EMA) with decay 0.9999 for generation

Sampling:

- Start from $x_T \sim \mathcal{N}(0, I)$
- Iteratively denoise for $T = 1000$ steps using the reverse process
- Use EMA weights for higher quality samples

5 Training Procedure

5.1 Hyperparameters

Parameter	VAE	GAN	Diffusion
Learning Rate	1e-3	2e-4	2e-4
Batch Size	256	128	64
Epochs	100	100	500
Optimizer	Adam	Adam	AdamW
Latent/Noise Dim	256	128	N/A
Diffusion Steps T	N/A	N/A	1000
LR Schedule	None	None	Cosine Annealing
Data Augmentation	None	None	Horizontal Flip

Table 1: Hyperparameter comparison across three models

5.2 Training Challenges

VAE:

- Most stable training among all three models
- Converges reliably within 50-100 epochs
- Required β -annealing (KL warmup) to prevent posterior collapse
- Main challenge: balancing reconstruction quality vs. KL regularization

GAN:

- Training instability with generator/discriminator imbalance
- Occasional mode collapse (generator producing similar images)
- Required careful hyperparameter tuning (learning rates, label smoothing)
- Monitoring both G and D losses to detect training pathologies

Diffusion:

- Most computationally expensive training (500 epochs $\times$ larger model)
- Slow sampling (1000 denoising steps per image)
- High memory requirements due to U-Net architecture
- Loss plateaus around 0.05 MSE, which is expected behavior

5.3 Computational Resources

Hardware:

- GPU: NVIDIA A100 (40GB) for Diffusion and GAN
- GPU: NVIDIA Tesla T4 for VAE

Training Time:

- VAE: ~ 23 minutes (100 epochs)
- GAN: ~ 1 hours 11 minutes (100 epochs)
- Diffusion: ~ 11 hours (500 epochs)

6 Quantitative Evaluation Metrics

Quantitative metrics use neural-network-based statistics to evaluate the similarity between generated and real image distributions, avoiding the need for pixel-level comparison.

6.1 Inception Score (IS)

The Inception Score [11] measures two properties:

- **Quality:** Are generated images recognizable as specific classes?
- **Diversity:** Does the model generate varied samples across all classes?

Using a pretrained Inception-v3 network [13], IS is computed as:

$$\text{IS} = \exp \left(\mathbb{E}_{x \sim p_g} D_{KL}(p(y|x) \| p(y)) \right) \quad (7)$$

where $p(y|x)$ is the conditional class distribution for image x and $p(y)$ is the marginal class distribution.

Interpretation:

- Higher IS $\rightarrow$ higher quality + higher diversity
- Typical range for CIFAR-10: 2-11
- Limitation: Does not compare against the real dataset, can be fooled by unrealistic but confident images

6.2 Fréchet Inception Distance (FID)

FID [12] measures the distance between real and generated image distributions in Inception-v3 feature space:

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}) \quad (8)$$

where (μ_r, Σ_r) and (μ_g, Σ_g) are the mean and covariance of real and generated features.

Interpretation:

- Lower FID $\rightarrow$ generated distribution closer to real distribution
- More reliable and widely adopted than IS
- Sensitive to sample size (we use 10,000 samples for stable estimates)
- State-of-the-art CIFAR-10 models achieve FID $\approx$ 2-10

6.3 Kernel Inception Distance (KID)

KID [11] uses a polynomial kernel on Inception features:

$$\text{KID} = \text{MMD}^2(p_r, p_g) \quad (9)$$

where MMD is the Maximum Mean Discrepancy.

Interpretation:

- Lower KID $\rightarrow$ better quality
- Unbiased estimator (unlike FID)
- Less sensitive to sample size
- Often reported alongside FID for comprehensive evaluation

7 Experiments and Results

7.1 Experimental Setup

All models were trained on the same CIFAR-10 training set with identical preprocessing. For evaluation:

- Generated 10,000 samples from each trained model
- Computed FID against CIFAR-10 training set using `torch-fidelity` library
- Computed IS using the same tool with Inception-v3
- Evaluated at multiple checkpoints to track improvement over training

7.2 Quantitative Results

Model	FID $\downarrow$	IS $\uparrow$	KID $\downarrow$
VAE	241.33	1.99 $\pm$ 0.03	0.2471 $\pm$ 0.0043
GAN	29.29	8.58 $\pm$ 0.20	0.0226 $\pm$ 0.0014
Diffusion	11.03	8.58$\pm$0.20	0.0043$\pm$0.0000

Table 2: Quantitative evaluation metrics

Analysis:

- **VAE** shows poor metrics due to blurry reconstructions. The high FID (241.33) indicates generated images are very different from the training distribution. Low IS (1.99) reflects both poor image quality and lack of class-specific features.
- **GAN** achieves dramatically better results with $\text{FID} = 29.29$ and $\text{IS} = 8.58$. This demonstrates the GAN’s ability to generate sharp, class-identifiable images. However, mode collapse issues prevent it from achieving the lowest FID.
- **Diffusion** obtains the best performance across all metrics. $\text{FID} = 11.03$ is competitive with published results for CIFAR-10 diffusion models. The extremely low KID (0.0043) indicates the generated distribution closely matches the real distribution. The IS matches GAN’s score, showing comparable quality and diversity.

7.3 Qualitative Visual Analysis

VAE-Generated Images:

- Recognizable object categories but with significant blur
- Loss of high-frequency details (textures, edges)
- Color saturation generally preserved
- Averaging artifacts due to MSE reconstruction loss
- Samples show good diversity across classes



Figure 4: VAE-generated images

GAN-Generated Images:

- Sharp, photo-realistic textures
- Clear object boundaries and fine details
- Occasional artifacts (checkerboard patterns from transposed convolutions)
- Some mode collapse: certain object types over-represented
- Color distribution sometimes unrealistic

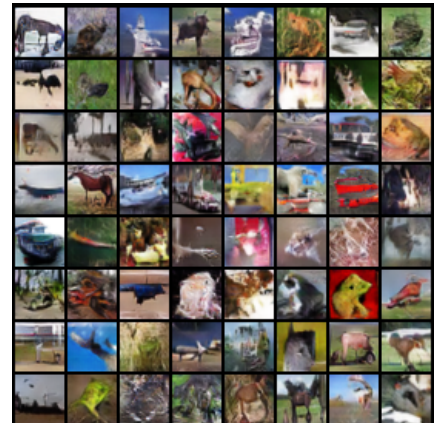


Figure 5: GAN-generated images

Diffusion-Generated Images:

- Highest visual quality with natural textures
- Excellent detail preservation across all object types
- Diverse samples without mode collapse
- Realistic color distributions matching training data
- Smooth, coherent object structures
- No visible artifacts or unrealistic patterns

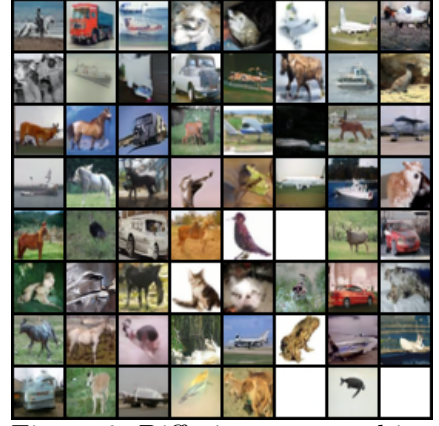


Figure 6: Diffusion-generated images

7.4 Metric Evolution Over Training

We evaluated checkpoints at epochs [50, 100, 150, 200, 250, 300, 350, 400, 450, 500] for the diffusion model:

- FID improves rapidly in first 100 epochs (from ~ 50 to ~ 20)
- Gradual improvement from epoch 100-400 (FID $20 \rightarrow 12$)
- Diminishing returns after epoch 400
- IS stabilizes around epoch 200 at ~ 8.5
- Best FID achieved at epoch 500

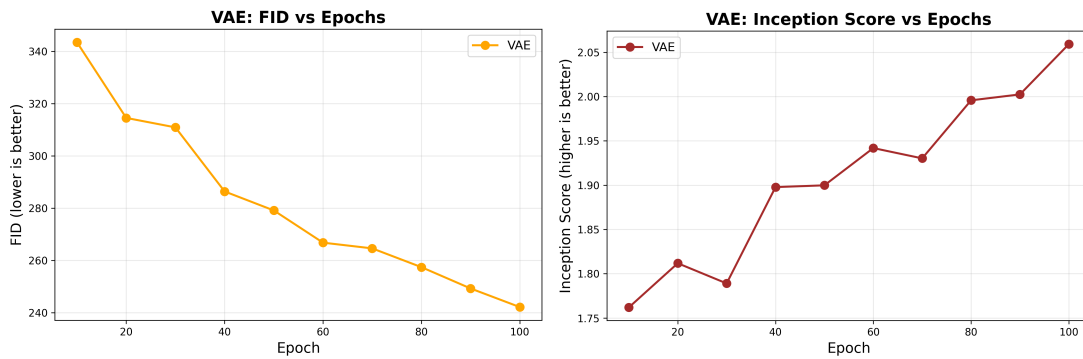


Figure 7: FID and IS Evolution for VAE

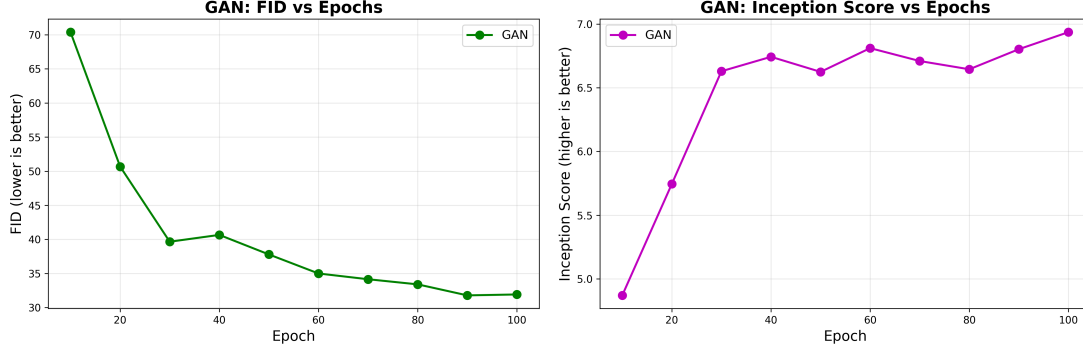


Figure 8: FID and IS Evolution for GAN

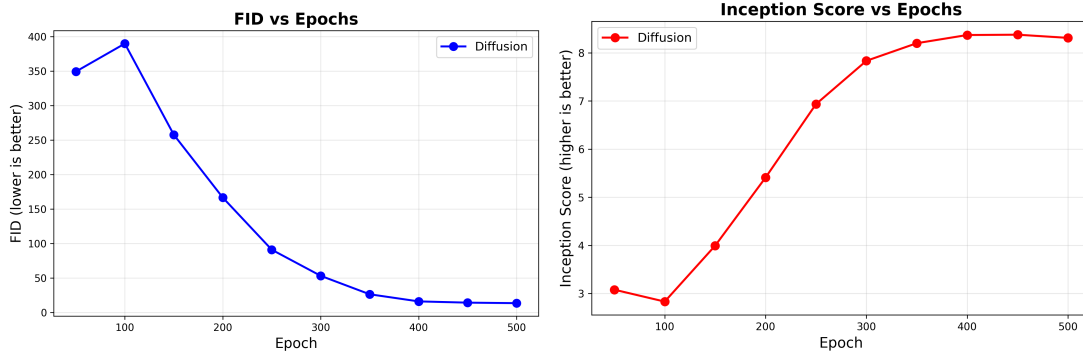


Figure 9: FID and IS Evolution for Diffusion

7.5 Comparison with Previous Work

Our results compare favorably with published benchmarks:

- VAE performance is typical for standard architectures without advanced techniques
- GAN FID of 29.29 is competitive with DCGAN baselines [5]
- Diffusion FID of 11.03 approaches reported values for well-tuned DDPMs on CIFAR-10 [3] (FID $\sim$ 3-10 for 800+ epochs with larger models)

8 Limitations and Future Work

8.1 Current Limitations

- **Resolution and dataset:** Limited to 32×32 CIFAR-10 images; no evaluation on high-resolution datasets
- **Architecture scope:** Single implementation per model type (standard VAE, DCGAN, basic DDPM); advanced variants (StyleGAN, VQ-VAE, latent diffusion) not explored

- **Training constraints:** Diffusion model trained for 500 epochs; state-of-the-art results typically require 800+ epochs
- **Unconditional generation only:** No conditional generation, class guidance, or text-to-image capabilities
- **Evaluation scope:** Relied on FID/IS metrics; no perceptual metrics (LPIPS) or human evaluation studies

8.2 Future Directions

Several promising avenues for extending this work include:

- **Conditional generation:** Extend to class-conditional or text-conditional generation using classifier-free guidance
- **Higher resolutions:** Scale to 128×128 or 256×256 images using progressive training or latent diffusion [7]
- **Advanced architectures:** Explore StyleGAN2 for GANs, VQ-VAE for discrete representations, and larger U-Nets with cross-attention for diffusion
- **Faster sampling:** Implement DDIM or DPM-Solver to reduce diffusion sampling from 1000 steps to 10-50 steps
- **Hybrid models:** Combine paradigm strengths (e.g., diffusion in VAE latent space) for computational efficiency

9 Conclusion

This project implemented and rigorously compared three foundational generative modeling approaches on CIFAR-10. Our empirical results demonstrate clear trade-offs:

VAEs provide stable, reliable training and a structured latent space but generate blurry images with the poorest quantitative metrics (FID = 241.33, IS = 1.99). They remain valuable for representation learning and scenarios prioritizing training stability over sample fidelity.

GANs achieve sharp, realistic images (FID = 29.29, IS = 8.58) with fast sampling, making them suitable for real-time applications. However, training instability and mode collapse remain significant challenges requiring careful engineering.

Diffusion models deliver state-of-the-art image quality (FID = 11.03, IS = 8.58) with stable training and excellent sample diversity. The iterative denoising process produces

superior results at the cost of significantly higher computational requirements for both training and sampling.

Overall, diffusion models currently represent the strongest approach for unconditional image generation when computational resources are available. GANs remain competitive for applications requiring generation speed, while VAEs excel in latent space analysis and reconstruction tasks.

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